

SKILL-BASED COURSE | 12 LEVELS | HTML / CSS / JS

Logic Building Course

Complete Web Page Design Prompt — All 12 Levels

"Logic is not a subject. Logic is how you think."

COURSE OVERVIEW

Course Philosophy

Logic Building is OTS's skill-based thinking course. It is not a mathematics class. It is not a coding class. It is a course about how to think clearly, solve problems step by step, identify patterns, reason through challenges, and make decisions logically.

The course has **12 Levels**. Level rank determines difficulty — not age or class. It is designed for complete beginners through to advanced learners.

The Four Gaps This Course Solves

Thinking Gap	Students react but do not reason.
Pattern Gap	Students memorise but do not see connections.
Problem-Solving Gap	Students give up when there is no obvious answer.
Confidence Gap	Students fear being wrong, so they avoid thinking.

GLOBAL DESIGN SYSTEM

Design System — Applied to All 12 Level Pages

Typography

Display / Headings: *Space Grotesk* or *Syne* — bold, geometric, modern.

Body: *DM Sans* or *Nunito* — clean, readable, warm.

Code / Logic blocks: *JetBrains Mono* or *Fira Code* — monospaced, techy.

CSS Color Variables

```
--bg-primary:#0D0D0D | --accent-primary:#00F5A0 (electric green) |  
--accent-secondary:#00D9F5 (cyan) | --accent-warm:#F5A623 (amber) |  
--accent-danger:#FF4757 (red) | --text-primary:#FFFFFF
```

Animation Principles

Page load: staggered fade-in from bottom (0.1 s delay per element).

Hover: transform translateY(-4px) + glow shadow.

Correct answer: green pulse. Wrong answer: red shake. Level complete: confetti burst.

Layout Rules

Mobile-first. Breakpoints: 320 px / 768 px / 1200 px. Max content width 1100 px. Section padding 80 px 24 px desktop, 48 px 16 px mobile.

Minimum touch target 48 px × 48 px. ARIA labels on all interactive elements. Minimum contrast ratio 4.5:1.

STRUCTURE

Level Architecture — Applied to Every Level Page

Every level page must contain these sections in order:

1. Level Header — number, name, tagline, difficulty badge, XP reward
2. What You'll Learn — 3–4 plain-language outcomes
3. Warm-Up Zone — low-stakes entry activity
4. Core Concept Section — main teaching content
5. Interactive Challenge — primary hands-on activity
6. Think Deeper — harder extension challenge
7. Real-World Connection — how this logic applies in real life
8. Level Badge + XP Unlock — completion reward
9. Navigation — Previous / Next level

THE 12 LEVELS

Level-by-Level Design Specifications

LEVEL 1
How Do I Think?Starter
100 XP*"Start at the very beginning — your brain."***Content & Interactions**

Introduce structured thinking. Sequences, patterns, daily logic. Morning routine drag-drop. Visual pattern sequences (shapes/colors). 'Make Your Own Pattern' builder. Real-world: traffic lights.

Page Design Direction

Full-screen hero with animated SVG brain. Floating thought-bubble cards. Soft particle background — dots connecting like neurons.

LEVEL 2
Yes or NoStarter
150 XP*"Every decision starts with a choice."***Content & Interactions**

Boolean logic — True/False, Yes/No, On/Off. Decision trees. True/False Blitz (10 Qs, 30 s). Decision Tree Builder (drag branches). 'Fix the Broken Logic' extension. Real-world: app logic.

Page Design Direction

Split-screen: LEFT=YES (green), RIGHT=NO (blue). Toggle switches as décor. Binary rain CSS animation.

LEVEL 3
Find the PatternExplorer
200 XP*"Patterns are everywhere. Can you see them?"***Content & Interactions**

Pattern recognition — visual, numerical, logical. AB to Fibonacci. Pattern Spotter grid. Number Patterns Lab (fill the blank, earn badges). Pattern vs Coincidence challenge. Real-world: Spotify, weather.

Page Design Direction

Geometric tessellation background. Puzzle-piece card entrance. Gold/amber accent.

LEVEL 4

Step by Step

Explorer

250 XP

*"Big problems are just small steps in the right order."***Content & Interactions**

Algorithms — plain-language intro. Recipe, directions, shoe-tying. Recipe Scramble drag-drop. Build the Algorithm (10 action blocks for Zara's morning). Spot the Bug extension. Real-world: Google Maps.

Page Design Direction

Step-counter UI. Blueprint/grid aesthetic. Each step card slides from the right.

LEVEL 5

What If?

Explorer

300 XP

*"Conditions change everything."***Content & Interactions**

IF/THEN/ELSE conditional logic. Nested conditions. Finish the Rule fill-in-blank. Flowchart Builder (drag IF/THEN blocks, test with 3 students). Nested Conditions predictor. Real-world: video games.

Page Design Direction

Dark purple-to-black gradient. Diamond design motifs. Animated flowchart lines. Scratch-inspired blocks without childishness.

LEVEL 6

Sort It Out

Challenger

350 XP

*"Order brings clarity."***Content & Interactions**

Sorting — Bubble Sort vs Selection Sort, visualised. No code required. Bookshelf drag-sort. Sorting Race animation (two methods side-by-side). Human Sort (swap-neighbours only). Real-world: Google search ranking.

Page Design Direction

Bold coral red. Animated bar charts. Racing aesthetic. Satisfying snap animations.

LEVEL 7

The Search

Challenger

400 XP

"Finding things fast is a superpower."

Content & Interactions

Linear vs Binary Search — logic of elimination. Hot or Cold warm-up (1–100 guess). Find It Faster side-by-side comparison (count steps). Design a Search scenario. Real-world: phone contacts.

Page Design Direction

Detective/mystery aesthetic. Spotlight effects on dark teal. Magnifying glass cursor.

LEVEL 8
Loop It

Challenger
450 XP

"Repetition is not boring. It's powerful."

Content & Interactions

Loops, repetition, exit conditions, infinite loops. Count with Me slider (even numbers, countdown). Loop Builder (30-student sticker scenario). Infinite Loop Trap — identify and fix. Real-world: YouTube recommendations.

Page Design Direction

Circular/orbital motion — elements orbit each other. Gold + dark. Spinning-out-of-control glitch for infinite loop.

LEVEL 9
Break It Down

Advanced
500 XP

"Decomposition: the secret weapon of great thinkers."

Content & Interactions

Decomposition — big to small. Smartphone component explode animation. Mind-map builder for canteen wait-time problem. Reverse Decomposition (8 tasks → identify the big system). Real-world: NASA moon landing.

Page Design Direction

Exploded-view blueprint aesthetic. Purple + dark. Cards literally break apart and reassemble.

LEVEL 10
The Rules of Reasoning

Advanced
600 XP

"Think clearly. Argue well. Decide wisely."

Content & Interactions

Deductive vs inductive reasoning, logical fallacies in plain language. Valid/Invalid Blitz (6 arguments). Fallacy Finder drag-label (straw man, false cause, hasty generalisation). Build the Argument (3-part). Real-world: law, science, daily decisions.

Page Design Direction

Sophisticated editorial. Magenta + white on near-black. Speech bubbles, scales, balance iconography.

LEVEL 11

Systems Thinking

Advanced

700 XP

"See the whole, not just the parts."

Content & Interactions

How parts interact — feedback loops, cause/effect, unintended consequences. Domino cause-effect chain sort. City Simulator (add cars → observe chain reactions). Design a Feedback Loop extension. Real-world: internet, economy.

Page Design Direction

Network/node aesthetic — glowing interconnected circles. Dark + sky-blue. Zoom in/out on network diagram.

LEVEL 12

The Logic Architect

MASTER

1000 XP

"Now you build the system."

Content & Interactions

CAPSTONE — apply all 11 concepts. Choose one complex problem (library system / football teams / home alarm). Complete a Logic Blueprint: problem, decomposition, algorithm, conditions, loops, system diagram. Gold confetti unlock + Logic Architect badge + certificate preview.

Page Design Direction

Gold and black — premium command-centre aesthetic. All previous level accent colours appear as background dots. Dramatic unlock sequence.

QUICK REFERENCE

Level Identity Quick Reference

Level	Name	Difficulty	Color Hex	XP Reward
Level 1	How Do I Think?	Starter	#00F5A0	100 XP
Level 2	Yes or No	Starter	#7BF1A8	150 XP
Level 3	Find the Pattern	Explorer	#F5A623	200 XP
Level 4	Step by Step	Explorer	#00D9F5	250 XP
Level 5	What If?	Explorer	#C77DFF	300 XP
Level 6	Sort It Out	Challenger	#FF6B6B	350 XP
Level 7	The Search	Challenger	#4ECDC4	400 XP
Level 8	Loop It	Challenger	#FFC300	450 XP
Level 9	Break It Down	Advanced	#A855F7	500 XP
Level 10	The Rules of Reasoning	Advanced	#F72585	600 XP
Level 11	Systems Thinking	Advanced	#00B4D8	700 XP
Level 12	The Logic Architect	MASTER	#FFD700	1000 XP

TECHNICAL SPECIFICATIONS

Global Technical Specifications

File Structure

```
logic-building/ ← Root index.html ← Course home / level selector level-01/ ... level-12/ ←  
One folder per level assets/css/ ← global.css + animations.css assets/js/ ←  
xp-tracker.js, interactions.js, confetti.js assets/fonts/ ← Preloaded font subsets
```

JavaScript Interactions Required

Drag and Drop: Native HTML5 drag API or lightweight library.

Answer Validation: Client-side, green/red feedback.

XP Tracker: localStorage to persist progress across sessions.

Progress Bar: Per-level and course-wide.

Confetti/Particles: CSS keyframes or lightweight canvas on level completion.

Performance

Total page weight per level: under 500 KB (no heavy frameworks).

Images: SVG preferred. PNG max 150 KB.

Font loading: font-display: swap.

Lazy-load illustrations below the fold.

Navigation

Sticky top nav: OTS logo + level number + XP counter.

Previous / Next level buttons at page bottom.

Breadcrumb: Home → Logic Building → Level X.

Full keyboard navigation support.

OTS-Specific Features

Local examples: Karachi traffic, school canteens, cricket team selection.

Urdu labels toggle for early levels.

Works on low-end Android at 2G speeds.

No login to explore — login only to save progress.

Teacher dashboard: level progress per student.

Parent view: weekly activity summary.